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ANNALS OF ONCOLOGY DRIVING INNOVATION IN ONCOLOGY

SPECIAL ARTICLE

ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP)

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Background: Large language models (LLMs) are rapidly being integrated into health care, with substantial implications for oncology practice. The European Society for Medical Oncology (ESMO) developed the ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP) to provide a structured framework and basic guidance for their safe and effective application in oncology.

Patients and methods: Between November 2024 and February 2025, a multidisciplinary group of 20 experts convened under the ESMO Real World Data and Digital Health Task Force. Using literature review and a Delphi consensus process, the panel defined three categories of LLM use in oncology: type 1 (patient-facing applications), type 2 [health care professional (HCP)-facing applications], and type 3 (background institutional systems). Consensus statements were developed for each type to provide basic practical guidance.

Results: ELCAP highlights opportunities such as improved patient education and symptom management, streamlined clinical workflows, and enhanced data processing. At the same time, it addresses challenges including data privacy, algorithmic bias, regulatory compliance, and the risk of unsupervised use. The framework emphasises human oversight, protection of patient privacy, and alignment with clinical and ethical standards. Patient-facing tools should complement, not replace, professional advice and should be embedded in supervised care pathways. HCP-facing and background systems may improve efficiency and decision support but require systematic validation, transparency, and continuous monitoring.

Conclusions: ELCAP provides a three-tier framework and basic practical guidance for LLM use in oncology. ESMO supports efforts to use this framework to improve patient care, but warns against unsupervised or unvalidated use.

Key words: large language model, AI, clinical decision making

INTRODUCTION

Large language models (LLMs) have made enormous technical strides from 2020 to 2025, and their evolution continues at an accelerated pace.[1-3] LLMs are artificial intelligence (AI) tools which perceive and produce natural language. From a technical point of view, LLMs work by

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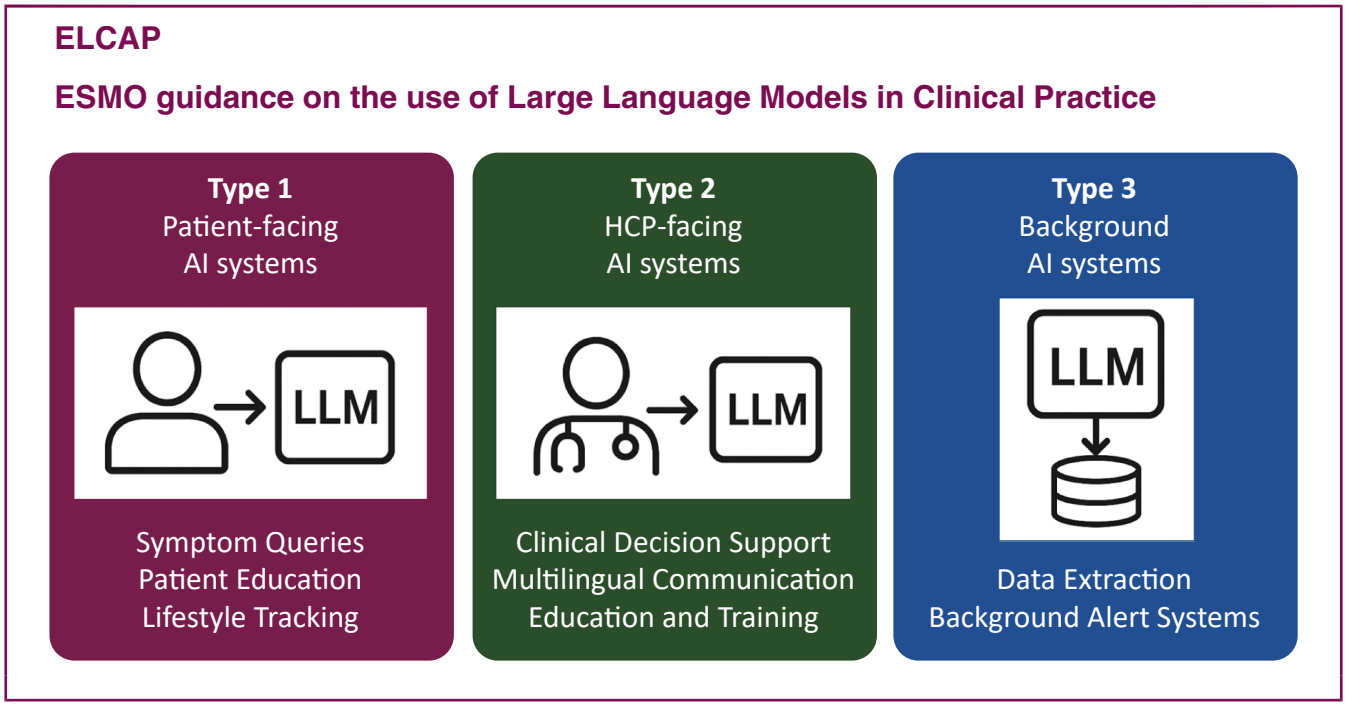
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GRAPHICAL ABSTRACT



Visual descriptions of the ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP) types. AI, artificial intelligence; HCP, health care professional; LLM, large language model.

breaking input text into smaller units called tokens, which can be words or sub-word fragments. Each token is converted into a numerical representation and passed through multiple layers of a neural network called a transformer. The model is trained to predict the next token in a sequence based on all the preceding tokens, learning patterns in language such as grammar, meaning, and context. By repeatedly carrying out this token-by-token prediction, the model can generate coherent text. Through training on extremely large datasets, the models learn human languages, but also high-level concepts, including medical concepts, and can subsequently carry out summarisation, conversation, question—answering, translation, and logical reasoning.[4] The potential of LLMs is currently being evaluated by health care professionals (HCPs), patients, and health care institutions alike, and other stakeholders. All of these groups are either planning to adopt or have already adopted LLMs for clinical or operational tasks.[5-7] Few other technologies in the history of humankind have achieved an uptake as fast as LLMs.

This technology has profound implications for the practice of oncology. The European Society for Medical Oncology (ESMO) acknowledges the need to provide guidance in this rapidly evolving field and to take on its role as a multi-stakeholder organisation to contribute to the swift, safe, and effective implementation of LLMs in clinical practice. Several recent publications proposed recommendations for LLM applications for health care organisations and clinicians to mitigate risks such as model drift, algorithmic bias, and data mismanagement while ensuring patient safety and effective clinical outcomes.[8-12] Consistent reporting in biomedical studies that leverage AI is in its infancy[13] and only recently the scientific and clinical communities began agreeing on standardised reporting tools to document LLM use.[10] There is substantial enthusiasm for clinical LLM adoption, growing interest for LLM explorations in clinical research, and yet a considerable unmet need to provide conceptual frameworks and guidance[14] on the use of LLMs specifically in oncology. With this aim, ESMO has convened a group of experts to develop the ESMO guidance on the use of Large Language Models in Clinical Practice (ELCAP). The key objectives of ELCAP are to provide a structured three-tier framework and basic practical guidance for oncologists, patients, and technology developers, among others, on how to evaluate and use LLMs in oncology, and to create a structured framework for their safe and effective implementation. This includes examining the current landscape of LLM tools in clinical practice and proposing strategies to manage their associated risks while maximising their potential benefits.

METHODOLOGY

Between November 2024 and February 2025, a group of 20 experts, including members of the ESMO Real World Data and Digital Health Task Force representing expertise in oncology, AI technology, biostatistics, clinical trial design, health care informatics, implementation science, digital health, and other

Table 1. Consensus statements issued by the ELCAP group

ID	Statement	*n*	Mean	Median	SD
Section 1: Statements regarding type 1 LLM applications—patients using LLMs					
1.2	Do phrase your questions clearly so that the LLM can provide accurate information.	20	7.60	8.00	1.62
1.3	Do review AI-generated advice with trusted resources or with your health care team before making important decisions.	20	8.50	9.00	1.32
1.4	Do keep an open dialogue with your doctor about any suggestions offered by the LLM.	20	8.35	9.00	1.01
1.5	Don't treat LLMs as a replacement for in-person consultations if you feel really unwell.	20	8.45	9.00	1.75
1.6	Don't share personally identifiable or sensitive health data in chat systems, unless these are made and approved for medical purposes and guarantee privacy.	20	8.40	9.00	0.73
1.7	Don't allow chatbot outputs to override recommendations from your physician.	20	8.80	9.00	0.40
Section 2: Statements regarding type 2 LLM applications—medical doctors and other health care professionals using LLMs					
2.1	Do verify all AI-suggested information by using guidelines, peer-reviewed studies, or expert consensus.	20	8.25	8.50	0.83
2.2	Do maintain human oversight, particularly for critical clinical decisions.	20	8.75	9.00	0.43
2.3	Do use LLMs to reduce paperwork (e.g. drafting referral letters) and focus more on patient care.	20	8.00	8.00	1.26
2.4	Do inform patients when AI systems are involved in their care.	20	7.55	8.00	1.50
2.5	Don't delegate final responsibility for patient communication, treatment plans, or diagnoses to an LLM.	20	8.70	9.00	0.56
2.6	Don't assume that AI outputs are automatically correct or unbiased; apply clinical judgement.	20	8.50	9.00	0.74
2.7	Don't upload sensitive patient data to external platforms without strict security and institutional and patient approval.	20	8.75	9.00	0.54
2.8	Don't let AI-generated suggestions overshadow direct communication with patients or colleagues.	20	8.75	9.00	0.70
Section 3: Statements regarding type 3 LLM applications—background AI in oncology					
3.1	Do ensure EHR documentation is complete and accurate to improve AI-generated outputs.	20	8.10	8.50	0.99
3.2	Do validate data extraction processes and confirm performance in both routine and new contexts.	20	8.15	9.00	1.15
3.3	Do implement continuous performance monitoring to detect potential errors, bias, or model drift.	20	8.50	9.00	0.81
3.4	Do train staff to recognise and question suspicious or unexpected AI findings.	20	8.45	9.00	0.74
3.5	Do establish an institutional governance process for background AI systems and promptly involve designated officers when necessary.	20	8.50	9.00	0.59
3.6	Don't replace human reviews with fully automated outputs for critical tasks like final diagnoses.	20	7.85	9.00	2.15
3.7	Don't rely on a single validation round; re-check models when workflows or data sources change.	20	8.30	9.00	0.84
3.8	Don't deploy AI solutions without evaluating their impact on existing clinical processes.	20	8.70	9.00	0.56

These items are recommended by the ELCAP expert panel. All statements which did not reach consensus (only Statement 1.1) are listed in Supplementary Table S1, available at https://doi.org/10.1016/j.annonc.2025.09.001.

AI, artificial intelligence; ELCAP, ESMO guidance on the use of Large Language Models in Clinical Practice; LLM, large language model; SD, standard deviation.

fields, including the patient perspective, convened in several meetings to discuss LLM applications in oncology. During the discussions, three types of LLM applications in oncology were defined. Type 1 LLM systems encompass patient-facing applications, type 2 LLM systems are used directly by HCPs, and type 3 LLM systems are institutional or 'background' systems. In the following sections, these three classes are briefly defined, and examples for existing and future AI solutions in this space as well as recommendations for their evaluation and use are provided. The ELCAP was divided into four project sub-groups, one for each type, and one cross-sectional group representing the patient, regulatory, and ethics perspective which was deemed relevant to all types alike. The groups carried out a literature review and identified published and ongoing efforts for LLMs in oncology and compiled them. In addition, all groups suggested statements on LLM use in oncology which they believed to provide important and practical guidance in this space. These statements were subsequently put to a Delphi procedure with all 20 experts, who evaluated them on a numeric scale from 1 to 9, with 1

indicating 'strongly disagree' and 9 indicating 'strongly agree'. A pre-specified criterion of a median $\geq$8 was used to indicate consensus of the group.[15] The results are displayed in Table 1.

ELCAP TYPE 1—LLM AI SYSTEMS USED BY PATIENTS

The current oncology care model struggles to address the complexity of modern cancer care in the face of workforce shortages and limited resources.[16] As a result, patients may experience gaps and disparities in informational and supportive care. A substantial part of their journey takes place outside the hospital setting, and this often requires them to seek guidance independently.[16]

ELCAP type 1 LLM systems are those which are directly used by patients or by their caregivers. These applications are directly patient-facing.[17,18] Such LLMs can potentially deliver personalised self-management advice, facilitate communication with HCPs, and empower patients to participate more actively in their treatment.[17,18] One example category is LLM-based symptom checkers,[19] or more generally, LLM-based applications giving health advice.[20] These tools can also be seen as digital guides, responding to patient queries about diagnoses, side-effects, and support services.[21] Thus, they could reduce uncertainty outside scheduled visits by offering on-demand information, potentially easing the stress of navigating complex treatment paths.

Many patient-facing applications of LLMs have been explored.[22] We will give several examples in the sections below. Overall, however, there is a striking lack of quantitative, empirical evidence regarding the clinical efficacy and safety of type 1 LLM systems. Formal validation studies remain scarce. Prospective randomised trials exist in psychotherapy and other medical fields, and these could be blueprints for oncology trials.[23] Such studies could evaluate patient-reported outcomes, engagement metrics, care pathway adherence, and reductions in unnecessary health care utilisation, or even survival endpoints. A common risk of current ELCAP type 1 systems is that they usually function as standalone solutions, disconnected from care pathways and therefore they are outside direct professional supervision with no clear protocols for escalating issues. This creates a pressing need for standardised scientific validation to address methodological inconsistencies and data privacy concerns, and to set appropriate metrics for assessing outcomes.[24,25] Another concern is the risk of hallucinations which can mislead patients or caregivers. At present, there are no validated safeguards to consistently prevent this phenomenon, and user detection of such errors may be challenging.

Examples of ELCAP type 1 AI systems

Patient-facing chatbots for symptom queries. A recent study evaluated a GPT-4-based chatbot for breast cancer patients based on the LLM GPT-4 to provide guidance on treatment-related toxicities.[26] Similarly, another GPT-4-based model was used to respond to patient messages in radiation oncology and was found to be accurate and empathetic.[27] These chatbots can access knowledge databases and, for example, reference hospital protocols to offer home-based symptom management advice. At the moment, LLM-based chatbots do not replace formal clinical advice but can provide additional detailed information to support patients. Reducing unnecessary emergency visits should only be considered a potential benefit when such tools are properly calibrated, clinically validated, and embedded in supervised care workflows. It is important to note that LLM-based symptom checkers and similar patient-facing chatbot systems are not currently approved as medical devices in most jurisdictions, including under the European Union Medical Device Regulation and United States Food and Drug Administration frameworks. We will discuss this in greater detail below. As such, their present role should be limited to exploratory or educational functions within well-defined, ethically approved research settings. While users are encouraged to critically review LLM-generated outputs and verify them with HCPs, we recognise that many patients may lack the health or digital literacy or access to resources necessary to do so reliably. Robust default safety features, plain-language outputs, and automatic escalation pathways for ambiguous or high-risk queries, particularly for vulnerable populations, could be used to address this.

Patient-facing clinical decision support systems. Some LLM-based platforms provide preliminary suggestions for possible diagnoses or therapies, or clinical trial enrolment.[19,28] In certain scenarios, patients might enter health data to receive an overview of relevant guidelines or diagnostic tests to discuss with their health care team. Some platforms also facilitate self-screening for clinical trials. Currently, these examples are drawn primarily from early-stage or proof-of-concept studies. These tools are not currently approved as medical devices, and therefore should not be used to guide clinical care. Another issue is that these tools could come with privacy vulnerabilities unless data are properly encrypted and safeguarded.

Patient education and information access. LLMs can offer tailored, patient-understandable explanations about diagnosis and treatment.[1,29,30] For instance, a newly diagnosed breast cancer patient might query an LLM, receiving plain-language summaries on staging, prognosis, possible treatment, and side-effect management.[31] In fact, many patients anecdotally seem to use general-purpose LLMs for exactly this purpose already. Compared with general-purpose LLMs, medical-specific LLMs could adapt their detail level according to user preferences, supporting patients before and after appointments. Such customisation is scalable, meaning it can be provided to patients at negligible marginal cost through the use of LLMs.[30] Key challenges are the need for appropriate tailoring of LLM advice and implementation at scale.

Lifestyle or symptom tracking and management. Patients often face evolving symptoms outside clinic hours.[32] LLM-based platforms could provide them with easier ways to

log symptoms, potential side-effects, medication adherence, and overall well-being. Compared with conventional symptom trackers, LLMs can enable unstructured data input which could reduce barriers for consistent logging. Such logging could trigger professional interventions upon trigger events. Also, LLM-based loggers could improve lifestyle and diet tracking. Formal research is still needed to confirm whether LLM-driven symptom tracking improves relevant clinical outcomes such as treatment tolerance and reduces hospital admissions.

Administrative support for patients. Patients are often required to manage multiple logistical and administrative tasks, including scheduling appointments, organising health records, and, in health care systems without universal coverage, navigating insurance processes and financial stress.[33] LLM-based assistants could improve these tasks by automatic scheduling, sending timely reminders, or automatically filling forms, thereby easing administrative burdens. While LLMs could enhance automation of routine administrative tasks, it is important to note that some of these functions, such as scheduling, could be achieved using simpler, rule-based tools that have existed for years. The limited adoption of such solutions reflects less a lack of technical capability than the persistence of institutional inertia, fragmented health IT infrastructure, and low prioritisation of administrative efficiency.

Virtual assistance and emotional support. Cancer diagnosis often triggers anxiety,[34] while professional mental health resources may be limited. LLM-based chatbots may provide basic coping strategies and self-care advice for distress,[23] with prompts advising users to seek professional help when symptoms worsen.[23] For some, simply having an instant, confidential, responsive interface after hours can alleviate concerns. Still, these chatbots must not substitute for professional mental health evaluation, and empirical evidence is needed to assess whether they improve outcomes as part of a broader care plan.

Practical guidance for ELCAP type 1 AI systems

Practical guidance for the use of patient-facing ELCAP type 1 AI systems was developed (Table 1). Seven out of eight statements reached a consensus in the ELCAP panel. The guidance outlines several key recommendations. Patients should phrase their questions clearly to ensure accurate LLM responses (Statement 1.2), while systematically reviewing any AI-generated advice with trusted resources or their health care team before making decisions (Statement 1.3). The guidance emphasises maintaining open dialogue with doctors about LLM suggestions, reflecting the complementary nature of AI tools (Statement 1.4). Critical safety boundaries are established by prohibiting the use of LLMs as replacements for in-person consultations during acute illness (Statement 1.5) and restricting the sharing of personally identifiable health data to approved medical platforms only (Statement 1.6). While LLMs may reduce informational burden, they should not displace clinician–patient interaction or increase provider workload through misaligned triage processes (Statement 1.7). Further guidance on this could be formally developed in the future, for example, stating that ELCAP type 1 AI systems should ideally not function in isolation but be embedded in human-guided clinical workflows.

ELCAP TYPE 2—AI SYSTEMS USED BY HEALTH CARE PROFESSIONALS

ELCAP type 2 AI systems are those which are directly used by oncologists and other HCPs. These systems are directly operated, or prompted, by human professional users. Thereby they are separated from ELCAP type 3 AI systems, which are discussed below, and which run autonomously in the background of electronic health record (EHR) systems. An ideal scenario is one in which ELCAP type 2 AI systems alleviate repetitive tasks, improve HCP performance and satisfaction, and at the same time improve clinical outcomes for patients.

Today, the local installation and use of LLMs in hospital routine is still in its infancy and faces many, mostly, practical challenges. Some of these factors are related to system-wide integration, while others pertain to data security and a multitude of usage types.[6] Similar to the adaptation of health care practices throughout the continuous technological developments of the last decades, HCPs should invest in obtaining AI literacy. This is an understanding of AI systems so that they can effectively use them and integrate them into their daily routines.[35] When carefully supervised, HCP-facing LLMs could improve care delivery, but transparency about limitations remains important.[26] In particular, ongoing debates focus on whether these models can introduce biases or oversights that might undermine the quality of care. Further concerns are automation bias and deskilling of physicians, effects on the patient–physician relationship, data privacy, lack of consistent contextualisation for local guidelines and regulations, as well as liability questions if negative outcomes ensue from AI-derived suggestions. The professional community should monitor and mitigate these potential risks by empirical, longitudinal evaluation. Importantly, ELCAP type 2 AI systems will only add meaningful value when their outputs are systematically, empirically evaluated, ensuring that they synergise with human-led clinical workflows.[36]

Examples of ELCAP type 2 AI systems

Clinical decision support. The main and primary use case of ELCAP type 2 AI systems is their use as clinical decision support software (CDSS). That is, a professional user asks a question related to patient care ("how should I treat this patient") and the LLM provides a response. Such systems have been evaluated in experimental settings for clinical cases and have shown promise to suggest potential diagnoses, point users to the correct guidelines,

or directly propose treatments.[24,37-40] When acting as CDSS, LLMs do not have to rely solely on the information acquired during training. Rather, they can access external information with information retrieval techniques such as retrieval-augmented generation.[38,41,42] Although questions of legal and professional liability here are complex, currently the clinician remains responsible for weighing these suggestions against their own judgement, patient preferences, local regulations, and resource constraints. Key challenges include biases, keeping systems up to date with clinical evidence, interoperability, and continuous improvements.

Multilingual communication. Other examples of ELCAP type 2 AI systems beyond CDSS include translation. Many oncologists care for patient populations that speak multiple languages and should deliver the same information to all of them. An LLM-based translation system can rapidly generate drafts of patient information leaflets or departmental memos, or even carry out real-time translation. There is some overlap with ELCAP type 1/3 systems in these applications, although we see translation systems which are operated by and prompted by HCPs as an ELCAP type 2 AI system. Key challenges are ongoing quality monitoring and ensuring cultural and other appropriateness.

Drafting documents. Oncologists and other HCPs spend a substantial amount of time drafting documents, such as medical reports, written communication with colleagues, payers, and administrators, and many other types of documents. A key capability of LLMs is that they can translate between different styles—one example for this is to translate from non-ordered, non-formatted, poorly legible notes into a well-formatted professional letter. Obviously, this process requires human oversight as long as LLMs have a non-negligible error rate but despite this, drafting professional documents with LLM support can save time. What is more, an LLM can be connected to an EHR (see ELCAP type 3 AI systems) and can automatically fetch important patient details from different information sources within an EHR. Key challenges are to ensure the quality and correctness of the resulting documents, but production-ready solutions seem within reach.

Education and professional development of health care professionals. Oncologists constantly educate themselves and are faced with an information overload due to the rapidly expanding literature and administrative complexity.[43] LLM-based search and summarisation tools can simplify that workflow, helping to identify relevant information, such as medical evidence or guidelines. An LLM might synthesise newly published data on anticancer therapies, linking to key references for further review. Physicians must still assess each source's reliability and relevance, but these tools can accelerate preliminary research. Studies are under way to determine whether such integrations ease decision making while maintaining safety.

Practical guidance for ELCAP type 2 AI systems

Practical guidance for the use of HCP-facing ELCAP type 2 AI systems was developed (Table 1). Eight out of eight statements reached a consensus in the ELCAP panel. These statements intend to provide authoritative guidance on how oncologists and other HCPs should approach and handle LLM-based systems. HCPs should verify all AI-suggested information against established guidelines, peer-reviewed studies, or expert consensus (Statement 2.1), while maintaining human oversight for critical clinical decisions (Statement 2.2). The guidance encourages using LLMs to reduce administrative burden, allowing more focus on patient care (Statement 2.3), and requires transparency about AI system use with patients (Statement 2.4). Professionals are explicitly warned against delegating final responsibility for patient communication, treatment plans, or diagnoses to LLMs (Statement 2.5), and must not assume that AI outputs are automatically correct or unbiased (Statement 2.6). Strict data protection should be observed: HCPs should not upload sensitive patient data to external platforms without robust security and explicit institutional and patient approval (Statement 2.7). Finally, AI-generated suggestions should never overshadow direct communication with patients or colleagues, ensuring that human interaction remains central to oncology care (Statement 2.8).

ELCAP TYPE 3—AI SYSTEMS IN THE BACKGROUND OF HEALTH SYSTEMS

A third type of LLMs in oncology is not immediately visible to patients or HCPs. We define these as ELCAP type 3 AI systems. These are not interacting directly with users but are deployed in the 'background' at health care institutions, operating unobtrusively within EHR systems. They improve processes rather than serving as hands-on usable tools. For example, these background tools can run automated data analysis, extract structured information from unstructured free text, and flag eligible trial candidates without requiring direct interaction from patients or clinicians.

The widespread adoption of EHRs in the 20th century has expanded the volume and availability of clinical data, creating opportunities for improved patient care and data-driven research. Nonetheless, daily users of EHRs also experience frustration and burnout, due to the increase in clerical tasks, volume of data, complex interfaces, availability of patient portals, and EHR messaging.[44] When LLMs are introduced into EHRs as ELCAP type 3 AI systems, a key aim should be for them to decrease, rather than increase, user frustration and to improve, rather than impair, productivity and safety. If these systems overpromise and prematurely replace expert human-based workflows, critical errors may go unnoticed. Ultimately, ELCAP type 3 AI systems could reduce repetitive tasks and streamline documentation, enhancing workflow efficiency, and ultimately contribute to a more supportive health care environment.

Examples of ELCAP type 3 AI systems

Data extraction and pooling. ELCAP type 3 systems can extract structured data from semi- or unstructured medical notes or reports, such as pathology or imaging reports, to extract key details such as cancer diagnosis/staging, biomarkers, treatment history and outcomes, treatment-related toxicity, symptoms, risk factors, patients' performance status, and comorbidities.[45-50] This can serve multiple downstream purposes. It can improve operational efficiency of clinical processes where structured information is required—for example, in filling tumour board documentation. It could also improve efficiency of administrative processes, for example, where structured information is required to initiate reimbursement. Additionally, it can be used for research processes, for example, when structured information is required to populate epidemiological databases which could save time for overburdened staff and expand opportunities for real-world evidence studies.[51] Risks arise when imperfect or incomplete documentation leads to inaccurate outputs, since LLMs rely on the data they are given and may hallucinate when data are absent. Thorough auditing and user training are essential to ensure that automated outputs do not propagate flawed data, disrupt processes, or, at worst, compromise patient safety. Ethical issues surrounding data privacy, possible unauthorised secondary uses of EHR content, and the risk of implementing automated systems that have not achieved adequate accuracy thresholds require careful oversight.

Background AI alert systems. Background systems could monitor care processes and alert oncologists to deviations from established guidelines, to potential adverse events to drugs, or to potential drug interactions. Such automated checking is difficult to implement with rule-based software systems but could be achieved with LLMs that excel at understanding context information. Such systems could catch errors early, but clinicians should retain final authority. There are potential dangers associated with such systems, however. Overdependence on alerts may dull critical thinking, and false positives can erode user trust. Genuine improvements require transparent governance, comprehensive training, and repeated performance evaluations. A balanced approach that pairs AI capabilities with human expertise is warranted.

Ambient AI scribes, for instance, can listen to patient—doctor conversations and automatically transcribe them. Several such AI systems are now on the market and promise to minimise documentation burdens and improve the clinician—patient experience.[52] They could ultimately be a stepping stone towards higher levels of AI automation in health care, in which AI does not just passively listen and transcribe, but also takes action, orders tests, or interacts with patients directly. While simple AI scribes are conceivably beneficial, such higher levels of automation require empirical evidence. They also blend the boundaries between ELCAP type 1, 2, and 3 AI systems. A special case of ambient AI scribes is tumour board transcription. Interdisciplinary tumour board discussions generate detailed management plans. An LLM might record these sessions and automatically produce concise summaries for team review. A human expert (similar to ELCAP type 2 applications) should check for accuracy before distribution, saving time on note taking. This approach could help standardise report formats, improving continuity of care.[53,54] However, it depends on reliable transcription, and any misinterpretations could affect treatment decisions, underlining the need for ongoing verification.

Clinical trial matching tools. Clinical trials in oncology have become increasingly specific to well-defined populations of patients. A specific task of high relevance in oncology is to screen patients for clinical trials and to screen clinical trials for patients. Such clinical trial matching is notoriously difficult to implement with rule-based computer systems, given that inclusion and exclusion criteria are highly complex, nested, conditional statements which are difficult to capture accurately. LLM-based systems have been proposed to solve this.[55-58] Such approaches[59] can automate a time-consuming manual process made by HCPs (similar to ELCAP type 2) and widen access to studies, including less common and rare tumour subtypes. Some studies also highlight how LLM-based solutions might enable patient self-screening for broader trial accessibility (similar to ELCAP type 1). Still, the system's success depends on complete, accurate records and careful oversight to confirm correct matches. Importantly, LLM-based clinical trial matching tools require validation in real-world cases with high complexity, not just in 'mock cases'.

Billing and reimbursement. LLMs could facilitate medical billing and coding by automating complex, error-prone tasks, extracting data for coding (International Classification of Diseases, ICD, Current Procedural Terminology, CPT), and validating compliance with payer requirements to prevent under- or over-coding. However, limitations of generalist models include inadequate domain-specific training, susceptibility to errors, and the need for precise alignment with regulatory standards and clinical nuances.[60] Integrating LLMs with knowledge datasets, agentic approaches could enhance accuracy.[61]

AI for research: identifying new predictors of disease evolution. Lastly, a final example of background AI systems is one that parses clinical and/or molecular data to identify patterns of prognostic or predictive values. Such clinical research applications of AI could be relevant in the setting of clinical trials, but also in the setting of real-world evidence. By analysing patient records in bulk, the system can highlight correlations that might otherwise remain unnoticed.[62] Proper validation is critical, however, to confirm that these insights generalise across various populations and practice settings. Hospitals often participate in data-sharing initiatives, though strict governance structures must prevent

biases or erroneous conclusions. If these models are rigorously tested, they could shape future research and ultimately inform more personalised treatment strategies. Other potential tasks, such as carrying out meta-analyses and literature reviews, are not discussed here.

Practical guidance for ELCAP type 3 AI systems

Practical guidance for the implementation of background ELCAP type 3 AI systems was developed (Table 1). All eight statements reached consensus among the ELCAP panel. These recommendations aim to inform health care institutions, clinicians, and administrators on the responsible use and oversight of AI tools integrated into EHR systems and backend workflows. Institutions should ensure that clinical documentation is complete and accurate, as LLM-generated outputs depend heavily on data quality (Statement 3.1). Before deployment, data extraction methods should be validated in both routine and novel clinical contexts (Statement 3.2). Continuous performance monitoring is essential to detect errors, bias, or model drift over time (Statement 3.3). Staff should be trained to critically evaluate AI outputs and question unexpected or suspicious results (Statement 3.4). Background systems should be governed by institutional oversight processes, with designated officers involved when issues arise (Statement 3.5). The guidance warns against replacing human review with fully automated decisions for critical tasks such as diagnoses (Statement 3.6) and mandates re-validation of AI tools whenever workflows or data sources are modified (Statement 3.7). Finally, institutions should assess the real-world impact of these systems on clinical processes before deployment to avoid unintended disruptions or safety risks (Statement 3.8).

Implementation considerations and regulatory heterogeneity

While the ELCAP framework provides a structured approach to categorise LLM applications in oncology, its practical implementation will depend on institutional context and regulatory constraints. Real-world experiences such as the internal deployment of GPT4DFCI at Dana-Farber Cancer Institute illustrate how institutions may begin to operationalise LLMs within existing infrastructures.[6] This effort combined technical safeguards (e.g. Azure-based architecture with single sign-on and audit logging), a dedicated governance structure, and phased user engagement through targeted training programmes. Importantly, use was limited to research and operational settings, explicitly excluding applications related to direct clinical decision making, in alignment with United States regulatory guidance. This use case corresponds primarily to ELCAP type 3 systems, though in practice, overlaps across tiers may emerge.[6]

This example also highlights a broader challenge: the heterogeneity of regulatory environments across regions. While frameworks such as General Data Protection Regulation (GDPR) (Europe), Artificial Intelligence and Data Act (AIDA) (Canada), and emerging national AI acts offer guidance, they differ in scope and enforcement. As such, any attempt to apply ELCAP principles across jurisdictions will likely require adjustments. Institutions may benefit from modular governance approaches that preserve core safeguards—such as transparency, validation, and human oversight—while allowing for regulatory adaptation. Explicitly acknowledging this limitation may help support the broader applicability of the framework across diverse health care systems.

CONCLUSION

The oncology community cannot ignore the potential benefits which AI technology can provide to cancer patients and should make additional efforts for these benefits to materialise. While the regulatory approval of LLMs lags behind their real-world use, all stakeholders (i.e. patients, clinicians, hospitals, LLM developers, electronic medical record vendors) should operate under a shared-responsibility model.[63] In parallel to these efforts, the next frontier is already in sight: agentic AI models, which, building on the success of LLMs, can fully automate complex workflows in oncology.[64-66] The current ELCAP considers assistive LLM systems, but does not address agentic AI systems which take action autonomously. Future efforts are needed to provide clear guidance on the safety and effective use of agentic AI systems in clinical practice.

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